700105 Simulation & Concurrency

UNIVERSITY of HULL

What to put in the lab books

- Join GitHub Classroom
  - https://classroom.github.com/a/IeDBPXHF
  - Ideally your work should be presented in markdown – just to make you aware of this for things like setting up a portfolio in GitHub
- Relevant code
  - You can also use this repository as your actual code repository
- Results of tests where appropriate
- Demonstrate that you understand what the code does
  - Description of the techniques used in the code
  - Include diagrams where appropriate
  - Relevant code
  - You can also use this repository as your actual code repository

Lab book 2 – Friday 21st Feb

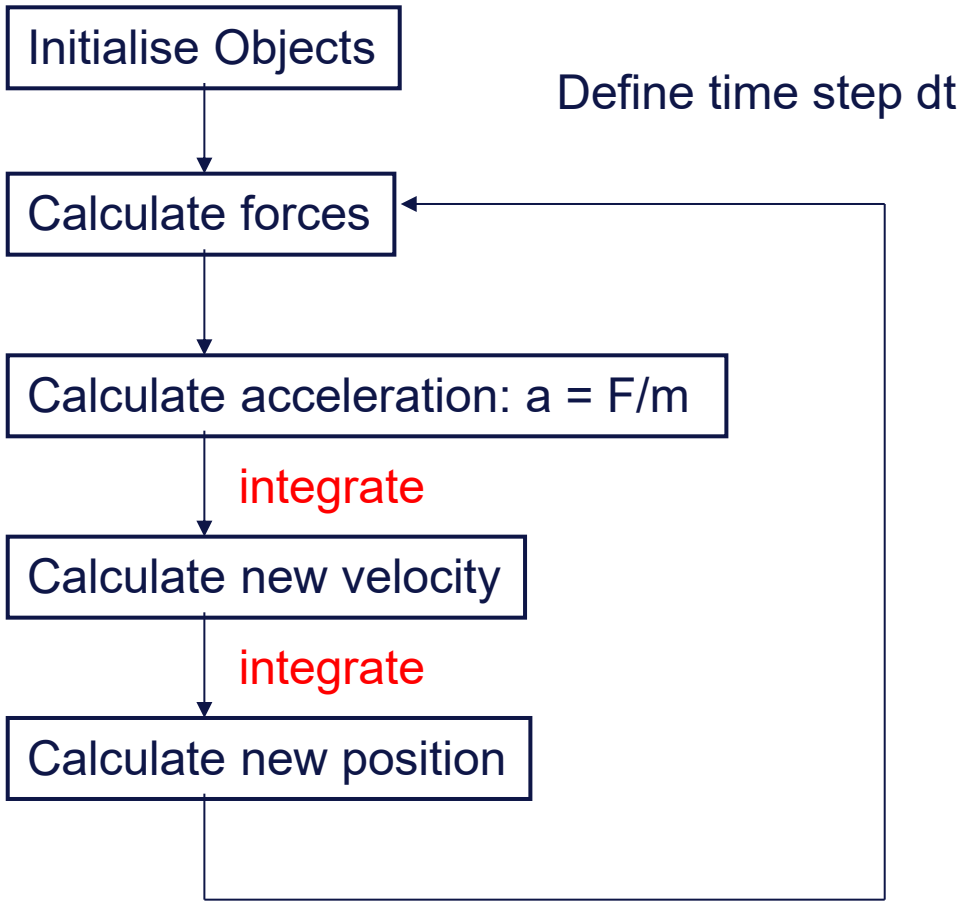
- **Q1 Be able to load and unload difference scenarios easily**
- **Q2 Be able to render primitive shapes**
- **Q3 Be ready to add simulation to the sandbox**

Story So Far

Indicative Content

- Updating State
  - Calculating forces
    - Gravity, Drag, Friction
  - Calculating new velocities and positions
    - Integration methods (Euler, Semi-Implicit Euler, Verlet, Runge-Kutta)
- Collisions - Geometric Objects (Shapes)
  - Primitives (point, line, line segment, plane, sphere, cylinder, capsule, cuboid, triangle, convex hull)
  - Methods (Inside-Outside, Intersects, Contains, SAT, GJK)
- Collision Response
  - Calculating new velocities or forces after collision
  - Conservation of momentum
- Reducing Calculations
  - Spatial Data Structures (uniform grid, quadtree, octree, kd-tree, BSP tree)

Simple Simulation Loop

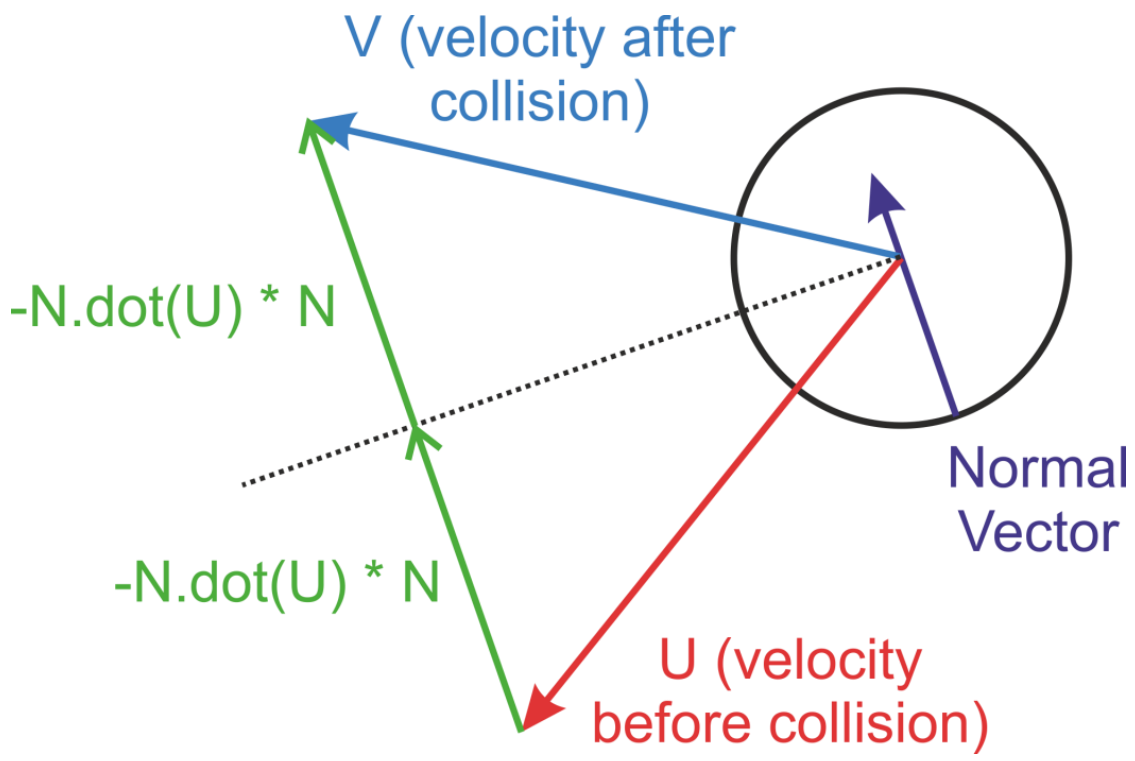


Note: No collision detection/response included at the moment

Collision Response

Calculating the reflected velocity

- V = U – 2 * N.dot(U) * N



Impulse

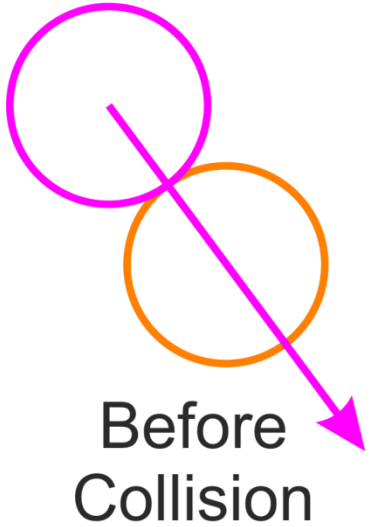
- Can either set the velocity to the new velocity
- Alternatively we can calculate the equivalent force given the fixed timestep (dt)
  - An ***Impulse*** is a the effect of force acting over time to change the momentum of an object
  - $J = F * dt$
  - $F = m * a = m * dv / dt$
  - $J = dt * m * dv / dt = m * dv$

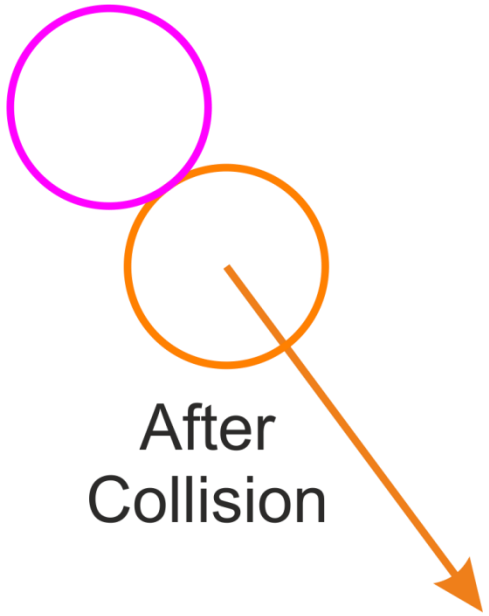
Rolling back positions

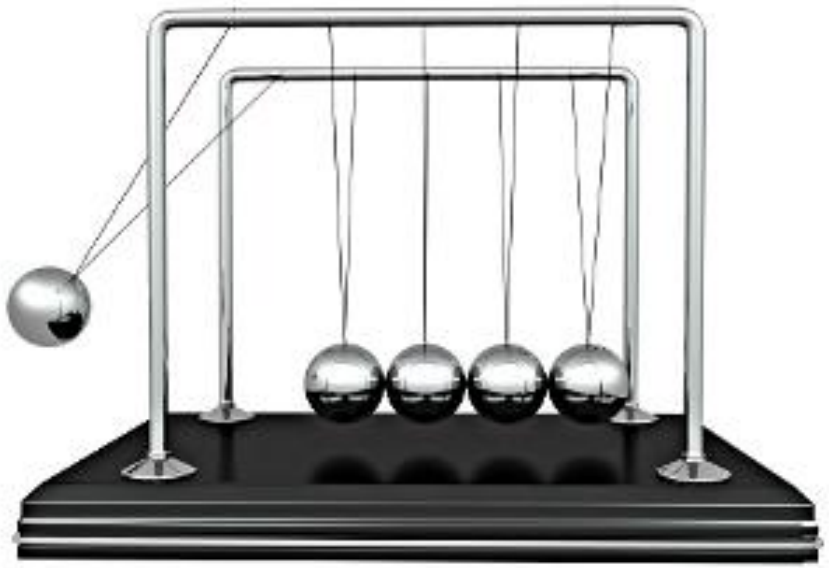
- For a stable system it is often good to ensure that no objects are left colliding one another
- One method is to store the old position of an object, and then roll that back if a collision is detected
- If more accuracy is required you might try moving the object half a timestep and then seeing if there is a collision

Head On Collisions

- Every action has an equal and opposite reaction (conservation of momentum)
- Total momentum before collision = total momentum after collision





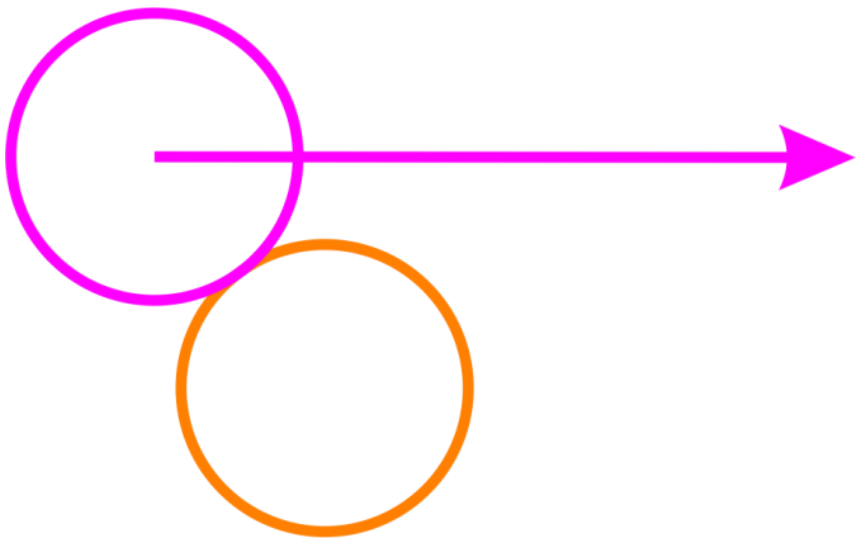


Direct collisions

- Can either set the velocity to the new velocity
- Alternatively we can calculate the equivalent force given the fixed timestep (dt)
  - An ***Impulse*** is a the effect of force acting over time to change the momentum of an object
  - $J = F * dt$
  - $F = m * a = m * dv / dt$
  - $J = dt * m * dv / dt = m * dv$
- We know dv so we can calculate the appropriate force to add given a reliable fixed timestep

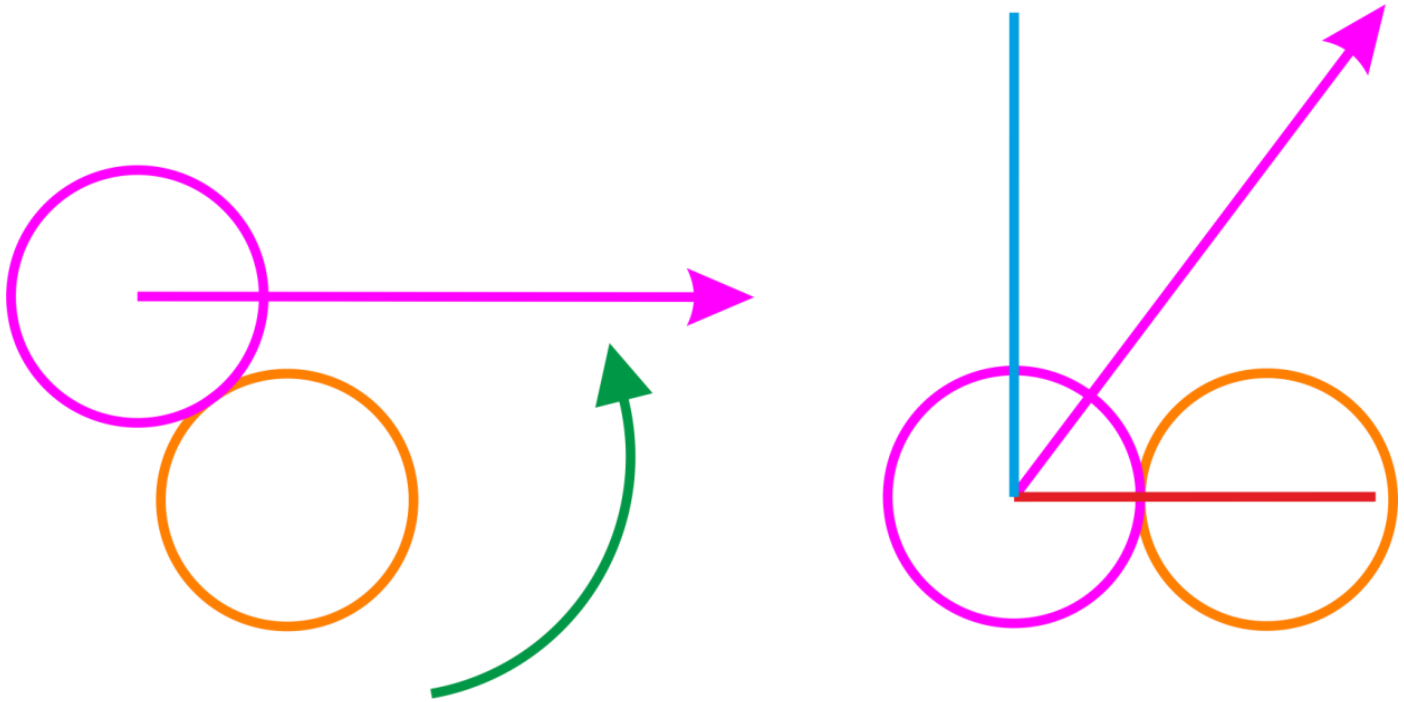
Collisions at an angle

- Momentum is conserved in the direction of the collision
- Momentum is retained perpendicular to the collision

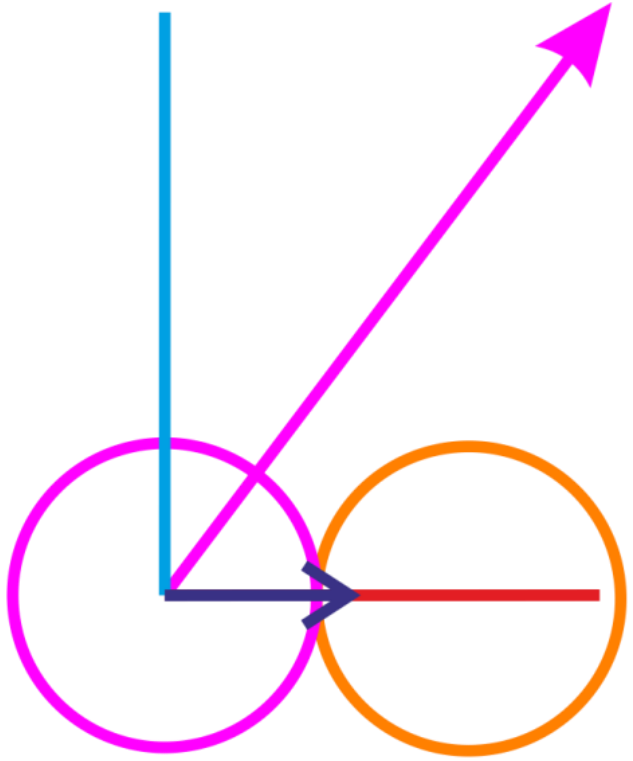


Collisions at an angle

- Turning a 2D problem that looks like a 3D problem into a 1D problem
- Here we consider the transfer of momentum (and velocity) in the direction of the collision (the red line) and parallel to the collision (the blue line)



Collisions at an Angle

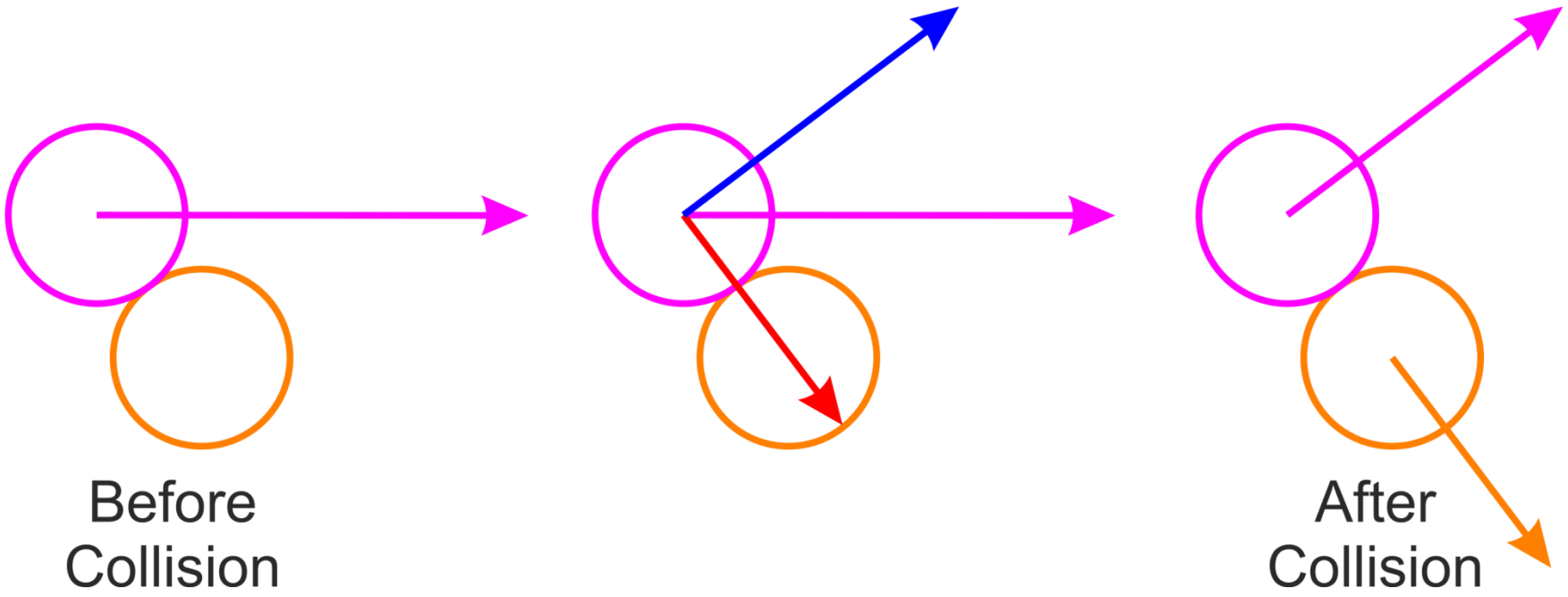


N = (P2 - P1).Normalize()

- Find direction of collision
- Parallel Component = V.dot(N)*N
- Perpendicular Component = V – Parallel Component

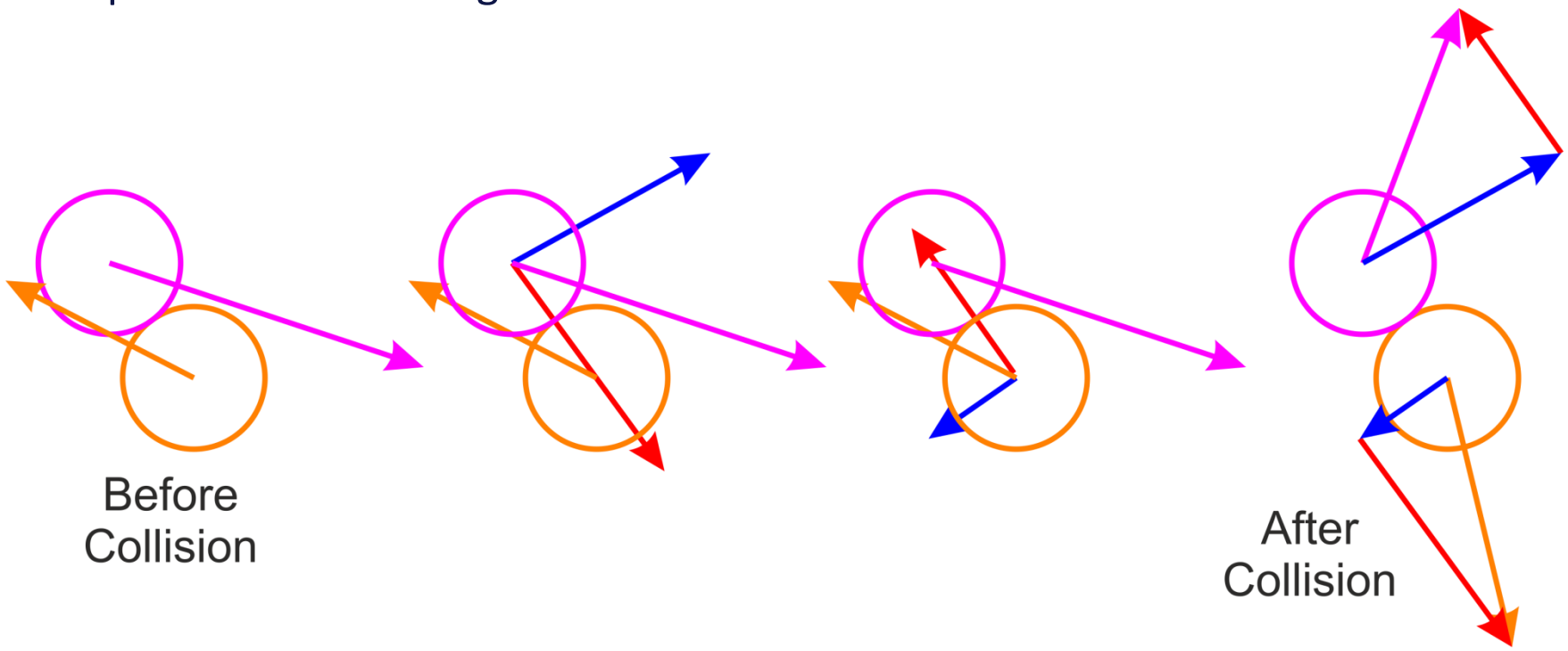
Collisions at an Angle

- Retain the perpendicular components and swap the parallel components
  - Only if particles have the same mass



Collisions at an Angle

- Retain the perpendicular components and swap the parallel components
  - Here both particles are moving



Time to “Code”

Canvas Workshop 4.1 – Start creating tests for direct collisions and collisions at an angle

Particles with different masses

Calculating & Storing Mass

- Mass = Volume x Density
  - Density is a property of a particular material
- Volume of Sphere $= \frac{4\pi r^3}{3}$
- Volume of a Cylinder = $\pi r^2 h$
- Because a = F / m we often store 1 / mass

Conservation of Momentum

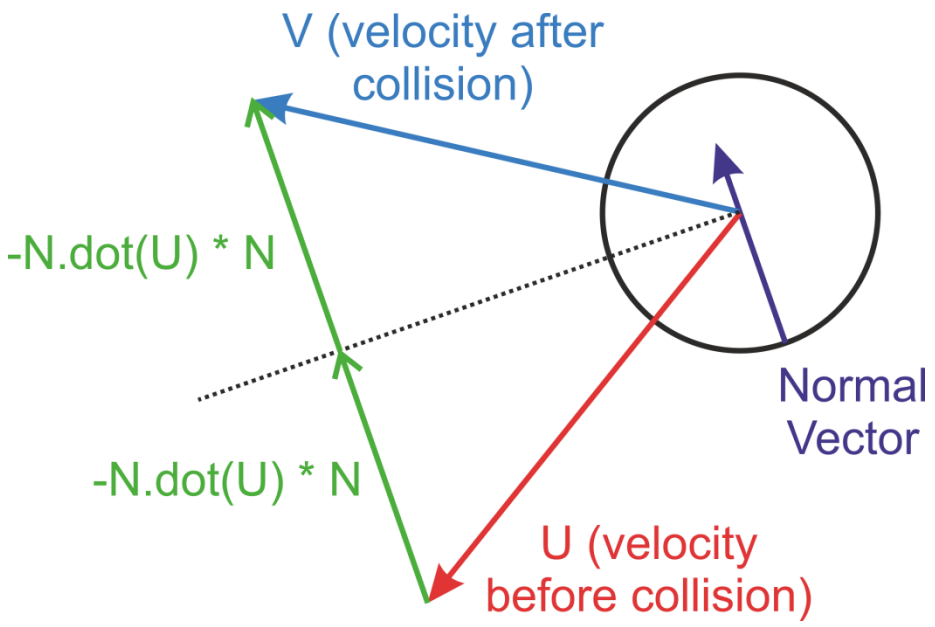
- Momentum = mass x velocity
- Momentum is conserved
- Energy is conserved
- $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) \times u_1 + \left(\frac{2m_2}{m_1 + m_2}\right) \times u_2$
- $v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) \times u_2 + \left(\frac{2m_1}{m_1 + m_2}\right) \times u_1$
- $u_1, v_1, u_2\ and\ v_2$ are velocity in collision direction

Momentum Examples – m_1 is much bigger than m_2

- $m_1 \gg m_2$
- $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) \times u_1 + \left(\frac{2m_2}{m_1 + m_2}\right) \times u_2$
- $v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) \times u_2 + \left(\frac{2m_1}{m_1 + m_2}\right) \times u_1$
- $v_1 = u_1$
- $v_2 = -u_2$
- Velocity is reflected!

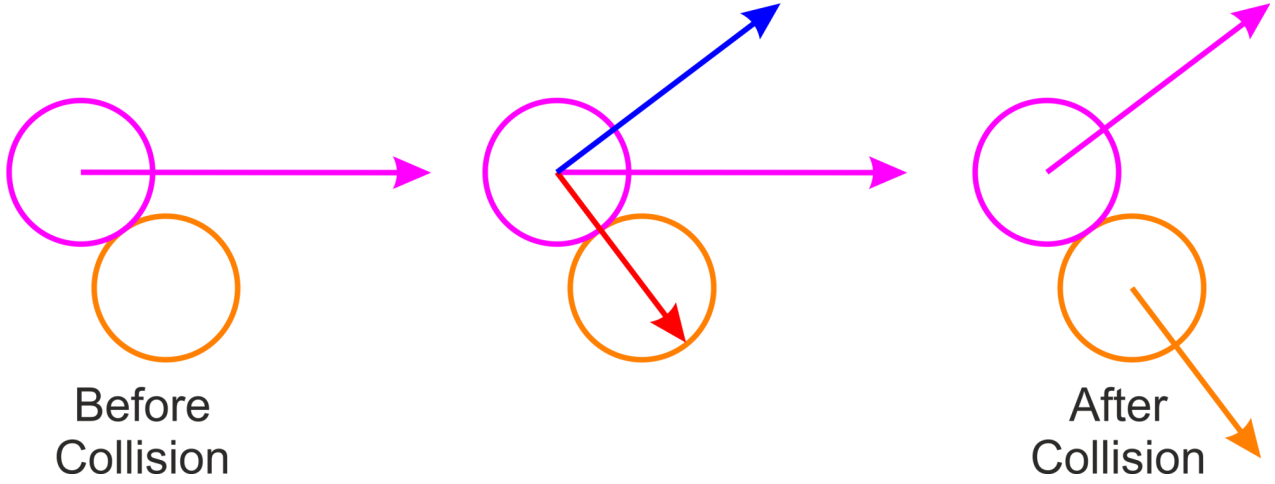
Sphere-Anything that doesn't move

- $v_1 = u_1$
- $u_2 = -u_2$ (in collision direction)



Momentum examples – m_1 is equal to m_2

- $m_1 = m_2$
- $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) \times u_1 + \left(\frac{2m_2}{m_1 + m_2}\right) \times u_2$
- $v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) \times u_2 + \left(\frac{2m_1}{m_1 + m_2}\right) \times u_1$
- $v_1 = u_2$
- $v_2 = u_1$
- Velocity is swapped!



Time to Code

Workshop 4.2

Elasticity

Adding Elasticity

- You can think of elasticity as “bounciness”
- Coefficient of Restitution e is a constant between 0 and 1 that is dependent on the two things that are colliding
  - Elasticity depends on all sorts of factors
    - (shape, material, internal structures)
  - Generally we with agree a value depending on the two objects that are colliding
- Coefficient of Restitution is applied to the change in velocities

Coefficient of Restitution

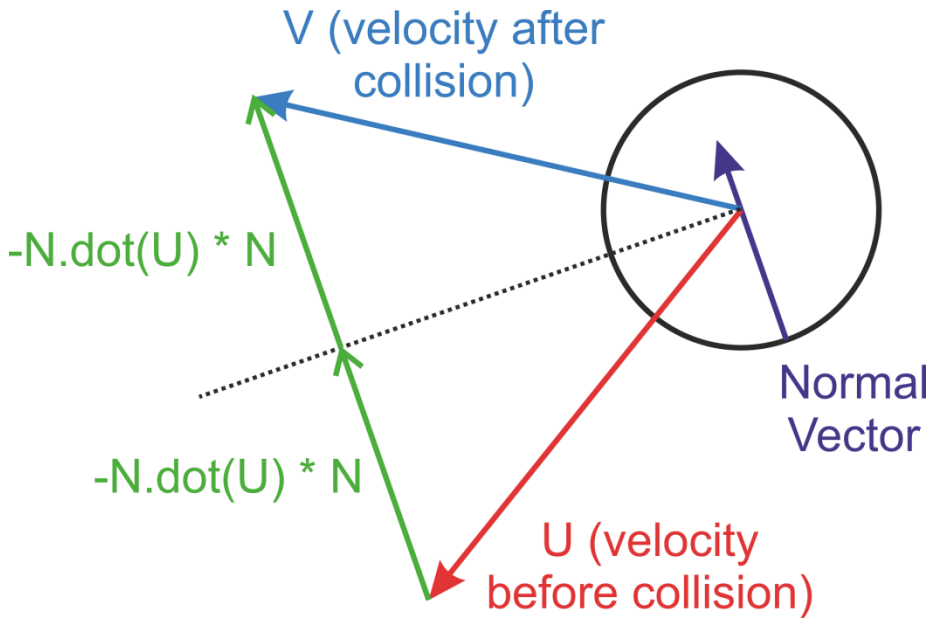
- Normally denoted by **e** (think of elasticity)
- The value of **e** is a function of the material and geometry of the colliding objects (2 or more)
- **e** defines a relationship between the velocities of the objects before and after a collision
  - Note that the velocities are measured along the line of impact (parallel to the collision)

Coefficient of Restitution

- **e** is the ratio of the relative velocities after the collision and the relative velocities before the collision
- $e = \frac{|V_1 - V_2|}{|U_1 - U_2|}$
- $\mathbf{V_i}$ is the velocity after collision for object i
- $\mathbf{U_i}$ is the velocity before collision for object I
- Elasticity affects the angles that objects rebound from after a collision
  - Elasticity reduces velocity in the collision direction

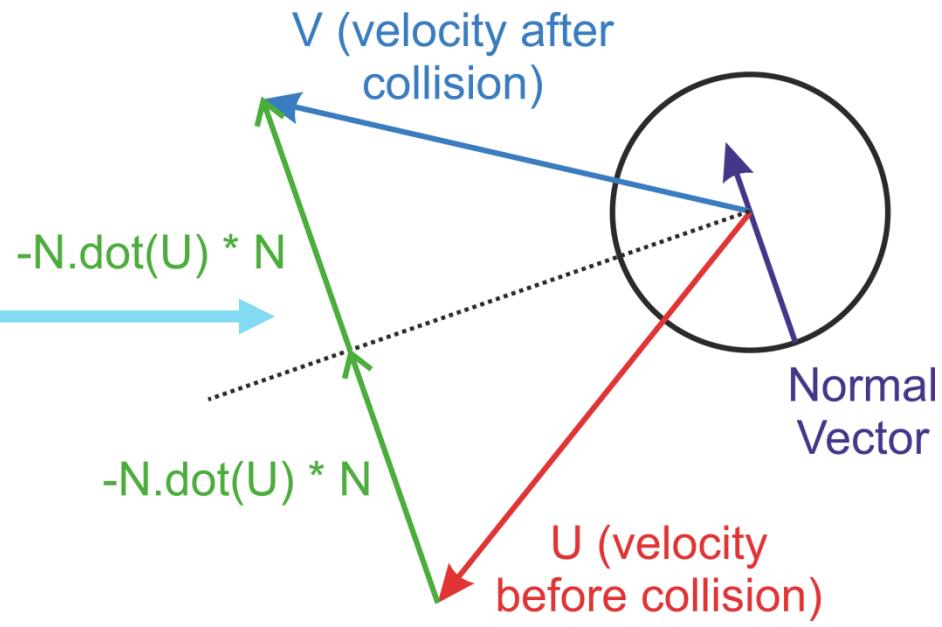
Collision with a fixed object

- In this simple case we only need to consider the velocity of the ball, as the floor does not move
- Elasticity only affects the component of the velocity that is parallel to the collision



Collision with a fixed object

- More formally Inelastic Collisions
  - $V = U - (1 + e) \times (N.U) \times N$



Note:

- When e = 0, the ball will slide on the ground
- When e = 1, no energy is lost

Collision with moving objects

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) \times u_1 + \left(\frac{2m_2}{m_1 + m_2}\right) \times u_2$$

$$v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) \times u_2 + \left(\frac{2m_1}{m_1 + m_2}\right) \times u_1$$

$$v_1 = \frac{m_1 u_1 - m_2 u_1 + 2m_2 u_2}{m_1 + m_2}$$

$$v_1 = \frac{m_1 u_1 + m_2 u_2 - m_2 u_1 + m_2 u_2}{m_1 + m_2}$$

$$v_1 = \frac{m_1 u_1 + m_2 u_2 + m_2 (u_2 - u_1)}{m_1 + m_2}$$

$$v_1 = \frac{m_1 u_1 + m_2 u_2 + m_2 \boldsymbol{e} (u_2 - u_1)}{m_1 + m_2}$$